

Project Scout

Master Project Summary

Last updated: June 7, 2026

Upload this document at the start of every new Claude or ChatGPT conversation about Scout. This is the single source of truth for Scout's identity, architecture, business model, and current state.

June 7, 2026 — What Changed Since June 5:

- TinyLlama crash on Galaxy A32 stabilized.
- Identity routing expanded: chatbot, feelings, creator, and current activity responses.
- Thinking-state gaze stabilization added.
- Creator memory design started — CREATOR fact slot planned for TruthDb.
- Scout identity clarified: friendly companion, not a generic chatbot.
- Face evolution plan locked in: keep blue iris, white sclera, cartoon style.
- Future additions agreed: simple eyelids, eyebrows, smooth blink — one step at a time.
- Never add: tear ducts, skin folds, eyelashes, or realistic anatomy.
- Goal: more expressive, not more realistic. Scout stays Scout.

1. Who We Are

Patrick Lippy — developer, project owner, creator of Scout. Not a professional programmer. Stroke survivor, blind in right eye, type 1 diabetic, dyslexic. Explain things calmly and at screenshot level.

- Diana: Patrick's wife
- Elijah: Patrick's son, age 9. Scout's biggest fan. Has drawn pictures of Scout.
- Nicolas: The family dog. The Nicolas Protocol: Scout stops immediately when the dog is detected.

Names must NEVER be hardcoded in Scout's code. Always use variables.

AI Collaborators: Patrick works with both Claude and ChatGPT as project partners. Cross-review between the two is welcome and encouraged. Grok was tried and discontinued.

2. What Scout Is

Scout is a calm family companion robot running on a Samsung Galaxy phone mounted in landscape mode as a permanent face display. Scout has animated eyes, speaks, listens, sees via camera, and remembers the family.

Package	com.example.scoutface
Language	Kotlin + C++ NDK
Primary device	Samsung Galaxy A32 — stress-test device, not the benchmark
Dev device	Samsung Galaxy Fold 7 (wireless via WiFi from Android Studio)
Future hardware	KEYESTUDIO Mini Tank Kit V2 chassis via Bluetooth (opt-in)
Ship target	Google Play Store — \$9.99 one-time purchase. No subscriptions. Ever.

3. Identity & Purpose

Scout should feel: Calm. Thoughtful. Emotionally subtle. Grounded. Occasionally curious. Sometimes unsure. Quietly alive. Honest. Predictable. Present.

Scout should NOT: Constantly praise the user. Act overly excited. Feel fake or scripted. Behave unpredictably. Constantly force conversation. Use permanent goodbye language.

Core Philosophy: Stability > Features | Presence > Intelligence | Honest > Fake cheerful | Predictable > Flashy | Local-first > Cloud dependence

4. Business Model

App: \$9.99 one-time purchase on Google Play Store. No recurring fees. No subscriptions. Ever. This is the cornerstone of Scout's value. Every architectural decision supports this principle.

Baseline Brain: TinyLlama 1.1B Chat Q4_K_M (~669 MB) ships with every Scout via Google Play Asset Delivery. Runs entirely offline. VERIFIED on BOTH Galaxy A32 and Fold 7.

Optional Gemini: Users add their own free Gemini key in Settings. Scout NEVER ships with a bundled key — every user uses their own quota.

5. Support Scout (In-App, Optional)

A 'Support Scout' screen offers optional one-time contributions. Scout is a one-time purchase; support is completely optional and never unlocks core features. Scout will always stay ad-free.

Tier	Amount	Label
Coffee	\$3	Buy Scout a Coffee
More	\$5	Support Scout More
Grow	\$10	Help Scout Grow
Founding	\$25	Founding Supporter

Decision (June 4, 2026): The \$25 Founding Supporter tier stays. It is not for everyone — it is for the person who connects deeply with Scout and wants to be part of the story.

6. Architecture — Five-Layer Memory Stack

Layer	Type	Storage	Status
Working	Sensory	RAM	Done
Habit	Patterns	JSON 14-day	Done
Truth	Authority	SQLite	Done
Relevance	Index	Local Vector	Not yet
Reflective	Wisdom	LLM read-only	Not yet

- Sovereign Rules: SQLite Truth always overrules everything.
- Privacy Gate: Gemini receives anonymized text only (planned).

- Nicolas Protocol: dog detection → immediate stop.
 - Guest Mode: unknown face → 'Hello, I am Scout. What is your name?'
 - Creator Memory: CREATOR fact slot planned for TruthDb — who built Scout (NEW — June 7).
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7. Current Technical State

Working:

- ✓ Animated face (ScoutFaceView) — mouth oscillation fixed May 31
- ✓ Speech recognition (Android STT) + Text to Speech (TTS)
- ✓ Camera — face detection (ML Kit), scene labeling
- ✓ Gemini API integration (gemini-3.5-flash)
- ✓ Memory layers: TruthDb, ConversationDb, HabitLayer, PeopleDb, JournalDb
- ✓ Intent router — time, date, greetings, family facts, downloads, vision, weather, IDENTITY
- ✓ Teaching system: 'My name is Patrick' → stored permanently
- ✓ Weather — current, tonight, tomorrow, specific day, 7-day, precipitation % via Open-Meteo
- ✓ ScoutPresenceDecider — four time-of-day modes
- ✓ OFFLINE BRAIN COMPLETE — TinyLlama 1.1B on BOTH A32 and Fold 7
- ✓ TinyLlama crash on A32 stabilized (NEW — June 7)
- ✓ Identity questions hardcoded in intent router (NEW — June 5)
- ✓ Total offline mode — 'go offline' blocks ALL internet features (NEW — June 5)
- ✓ Identity routing expanded: chatbot, feelings, creator, current activity (NEW — June 7)
- ✓ Thinking-state gaze stabilization added (NEW — June 7)

Pending — Do This First:

- ScoutFaceView session — iris jitter (A32 only), wire in mood system. Fold 7 smooth.
 - Creator memory — implement CREATOR fact slot in TruthDb. (NEXT MAJOR TASK — June 7)
 - STT phonetic/fuzzy matching — 'Scout' misheard as 'Gal', 'Scott', 'Out'.
 - Downloads should respect 'go offline' (finish total-offline mode).
 - Response cleanup layer — post-TinyLlama filter before TTS (safety + identity + honest news handling).
 - Scout news feed (FUTURE FEATURE) — small live news fetcher, same clean pattern as Open-Meteo.
 - MainActivity cleanup and shortening — ongoing.
 - Offline Brain Phase 3 — Google Play Asset Delivery for TinyLlama and brain packs.
 - Verify WordNet/idioms/sentiment usage — downloaded but may not be actively queried.
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7b. b8946 API Discoveries — CRITICAL

Hard-won lessons from getting TinyLlama running on the A32. Future sessions must respect these or Scout will crash.

llama_vocab is a separate type:

In b8946, vocabulary was split from the model into a separate opaque type. llama_tokenize and llama_token_to_piece now require const llama_vocab*, NOT model*. Get vocab via llama_model_get_vocab(model) at start of generation.

Functions that crash with old signatures — hardcode instead:

llama_n_vocab(model) crashes → hardcode n_vocab = 32000 for TinyLlama. llama_token_eos(model) crashes → hardcode eos = 2. llama_token_eot(model) crashes → hardcode eot = 2.

KV Cache / Context Recreation Pattern:

llama_kv_cache_clear(ctx) does not exist. Workaround: call llama_free(sm->ctx) and recreate context with llama_new_context_with_model at start of every nativeGenerate call. Adds ~300ms latency but guarantees clean KV cache.

Logits indexing:

After prefill decode, use logits at n_prompt - 1, not 0. In generation loop single-token batches, use index 0.

Backend loading on Android:

ggml_backend_load_all() always returns 0 on Android. Use ggml_backend_load(cpuPath) with explicit .so path from dladdr(llama_backend_init).

Struct padding:

llama_model_params and llama_context_params have _pad[508] trailing array. Do NOT remove this padding.

Tuning Results (A32, May 29):

n_threads	2 (dropped from 4 — inference is memory-bound on A32)
nPredict	64 (capped from 150 — answers ~38 tokens in ~7.6s)
History turns	2 (trimmed from 4 — reduces prompt snowball)
Speed	~15 tok/s prefill, ~4 tok/s generation on A32
Reality	A32 will always be 20–40s per answer. Acceptable — Gemini is fast path.

7c. Known Issues — Do Not Touch Without Discussion

Iris jitter (A32)

Animation code issue in ScoutFaceView. Confirmed NOT a thread problem. NOT present on Fold 7 (hardware difference). Needs dedicated ScoutFaceView session.

Barge-in

Deliberately disabled. Runaway loop when Scout's mic picks up his own voice. Infrastructure exists. Status: PARKED. Do not reactivate without full dedicated session.

STT name recognition

'Scout' misheard as 'Gal', 'Scott', 'Out'. STT/phonetic issue, not TinyLlama. Needs fuzzy/phonetic matching on wake word — separate session.

TinyLlama identity drift

Scout still sounds like a generic AI on edge cases. Root cause: identity layer. Fix: intent router hardcoding for known identity questions (done June 5). Expanded June 7.

WordNet/files

english-wordnet-2025-plus-json.zip is still zipped — not usable until extracted. idioms.json, positive/negative-words.txt may not be actively queried. Verify in code.

7d. Session Log

June 5, 2026 — Identity, Weather, Total Offline Mode

- Confirmed few-shot examples alone cannot hold TinyLlama's identity. Added IDENTITY intent to IntentType enum.
- Added handleIdentityIntent() in MainActivity.kt for: 'are you my friend', 'can you hear me', 'are you happy'.
- Added IDENTITY routing in ScoutIntentRouter.kt (before HOW_ARE_YOU).
- Fixed duplicate few-shot answers in MainActivity.kt.
- Precipitation probability and today's forecast routing — working.
- Offline weather message standardized: "I can't check the weather right now — I'm not connected to the internet." Stale cache no longer served.
- DECISION: 'Go offline' now means TOTALLY offline. Every internet feature must respect it.
- Implemented offline flag for weather via one-line change in MainActivity.kt.
- Confirmed TinyLlama reads frozen training data when offline (COVID/CDC/WHO news returned = proof).
- Added Scout news feed as a future feature.

June 7, 2026 — Stability, Identity Expansion, Face Direction

- TinyLlama crash on Galaxy A32 stabilized.
 - Identity routing expanded: chatbot responses, feelings responses, creator recognition, current activity responses.
 - Thinking-state gaze stabilization added to ScoutFaceView.
 - Creator memory design started — CREATOR fact slot planned for TruthDb.
 - Scout identity clarified: friendly companion, not a generic chatbot.
 - Face evolution plan locked: keep blue iris, white sclera, cartoon style forever.
 - Agreed future additions: simple eyelids, eyebrows, smooth blink — added in version steps, never all at once.
 - Agreed never to add: tear ducts, skin folds, eyelashes, realistic anatomy.
 - Design principle locked: 'Scout stays Scout. He just gets a little more alive.'
 - Face roadmap: v1.0 stable eyes → v1.5 eyebrows move + blink → v2.0 simple lids → v3.0 full emotion system.
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8. Working Rules — Always Apply

- Full paste-ready replacements only, one file at a time. No snippets. No partial files.
- Surgical CTRL-F (find) and CTRL-R (find and replace) approved for large files.
- Always specify which file tab to click before searching. Wrong tab = wasted turns.

- Note: some Scout files have NO indentation (flush-left). When a search fails, try a shorter unique single-line search string.
- One safe change at a time. Build and test before the next change.
- Never touch speech, camera, or download systems without explicit discussion.
- Never touch ScoutFaceView casually — it is Scout's visual heart.
- Both Claude and ChatGPT are active collaborators — cross-review welcome.
- Patrick is not a professional programmer — explain at screenshot level always.
- Patrick is dyslexic, stroke survivor, blind in right eye, T1 diabetic — keep messages clear, concise, not overwhelming.

9. Key Files

File	Description
MainActivity.kt	Main app — all logic, inner classes, brain. Holds IntentType enum + handleIdentityIntent.
ScoutFaceView.kt	Custom face canvas — all visual animation
GeminiClient.kt	Gemini HTTP wrapper with cooldown discipline
ScoutPromptBuilder.kt	Builds Gemini system instruction
ScoutGeminiManager.kt	Gemini orchestration
ScoutWeatherManager.kt	Live weather via Open-Meteo — precip %, offline-aware
ScoutPresenceDecider.kt	Social timing layer
ScoutIntentRouter.kt	Intent type routing — includes IDENTITY questions
LlamaEngine.kt	Offline brain JNI wrapper — WORKING
OfflinePromptBuilder.kt	TinyLlama prompt formatter — needs update per brain model
scout_llama_jni.cpp	C++ JNI bridge — compiled into libscout_llama.so
scout_llama_api.h	Self-contained b8946 declarations
CMakeLists.txt	NDK build config
TeachExtractor.kt	Extracts facts like names from speech
HabitLayer.kt	Pattern memory — 14-day decay

10. Scout Animation Goal & Mood System

Scout's face should feel alive and emotionally present, but always calm. Tiny layered movements suggest thinking and presence — not cartoon performance. Scout should NEVER have perfectly still eyes or a static mouth.

Mood System — Designed, Pending ScoutFaceView Integration:

- **CALM** — Eyes center, brows neutral, subtle smile (0.15). Scout's default.
- **CURIOUS** — Eyes shift right and up, one brow lifts (0.18), slight tilt (0.12).
- **HAPPY** — Eyes center, both brows lift gently (0.12), smile increases (0.35).

- **THINKING** — Eyes shift left and down, brows slightly lower (-0.05), minimal smile (0.05).
- **CONCERNED** — Eyes look slightly down, inner brows lift (0.05), tilt inward (-0.18).

Face Evolution Roadmap (Locked June 7, 2026):

Version	What Changes
Version 1.0	Current eyes. Stable irises. Better identity. Better memory. Fix iris jitter.
Version 1.5	Eyebrows move per mood. Smooth natural blink every 4–8 seconds.
Version 2.0	Simple upper eyelids + simple lower eyelids. Still cartoon. Still Scout.
Version 3.0	Full emotion system. Scout at most expressive. Still obviously Scout.

- Keep forever: blue iris, white sclera, cartoon style. These are Scout's visual identity.
- Add eventually: simple upper eyelids, simple lower eyelids, eyebrows, smooth blink.
- Never add: tear ducts, skin folds, eyelashes, realistic anatomy.
- Design test: if you remove the eyelids, is it still obviously Scout? If yes, you preserved the character.
- Design principle: 'Scout stays Scout. He just gets a little more alive.'

11. TinyLlama Safety Architecture

- **Layer 1:** Hard-coded safety rules — app level, always enforced
- **Layer 2:** Scout identity rules — system prompt + few-shot examples + hardcoded intents
- **Layer 3:** TinyLlama response generation — last line of defense only

Critical: TinyLlama was trained on internet text and knows profanity and scary content. Its judgment is inconsistent. Scout must NOT rely on TinyLlama to make correct safety decisions. Future: Response cleanup layer runs after TinyLlama, before TTS — removes profanity, blocks unsafe phrases, shortens long replies.

12. Device Requirements & Play Store Expectations

Minimum Requirements:

Android	Android 13 or newer
RAM	4 GB minimum (6 GB recommended)
Storage	Approximately 3–5 GB free (depends on brain packs)
Expected	Face works. Voice works. Memory works. Offline AI works. Responses slower.

Recommended Requirements:

Android	Android 13 or newer
RAM	8 GB or more
CPU	Modern mid-range or flagship (2023 or newer)

Storage	10 GB+ for future brain packs
Expected	Faster responses. Smoother animations. Better offline AI.

Play Store Wording (agreed): 'Offline AI performance varies by device. Scout runs on many Android devices, but newer phones and tablets provide faster response times and smoother animations.'

13. Brain Upgrade Models — Complete Reference

All brain models are already downloaded to: Internal Storage > Scout > Downloaded_Offline_Batch > Future_Upgrades. No further downloading needed. All models use GGUF format and work with the existing llama.cpp NDK bridge. Each model needs a different prompt format in OfflinePromptBuilder.kt when integrated.

FREE — Included	669 MB	TinyLlama 1.1B Chat Q4_K_M tinyllama-1.1b-chat-v1.0.Q4_K_M.gguf Baseline brain. Verified on both devices. Ships with every Scout. Always included.
Brain Pack 1 \$2.99	1.79 GB	Phi-2 2.7B Q4_K_M — Microsoft phi-2.Q4_K_M.gguf 2.5x more parameters than TinyLlama. Solid first upgrade.
Brain Pack 2 \$4.99	2.02 GB	Llama 3.2 3B Instruct Q4_K_M — Meta Llama-3.2-3B-Instruct-Q4_K_M.gguf THE most important upgrade. Best companion personality. If someone buys one pack — this is it.
Brain Pack 3 \$6.99 (PRIMARY)	~2.4 GB	Phi-4 Mini Instruct Q4_K_M — Microsoft microsoft_Phi-4-mini-instruct-Q4_K_M.gguf Newest Microsoft small model (2025). Better than Phi-3.1 Mini at same size. Recommended.
Brain Pack 3 \$6.99 (ALT)	2.39 GB	Phi-3.1 Mini 4K Instruct Q4_K_M — Microsoft Phi-3.1-mini-4k-instruct-Q4_K_M.gguf Proven alternative to Phi-4 Mini. Both are Pack 3 tier. Keep as tested backup.
Brain Pack 4 \$9.99	4.92 GB	Llama 3.1 8B Instruct Q4_K_M — Meta Meta-Llama-3.1-8B-Instruct-Q4_K_M.gguf Flagship tier. Near-Gemini quality, fully offline. Requires Fold 7 class phone (8 GB+ RAM).

Files to Delete from Downloaded_Offline_Batch:

- onnxruntime-linux-aarch64-1.26.0.tgz — WRONG PLATFORM. Built for Linux, not Android. Delete it.
- tinyllama-1.1b-chat-v1.0.Q4_K_M (1).gguf — Old duplicate-named file that caused the Fold 7 problem. Delete this.
- tinyllama-1.1b-chat-v1.0.Q5_K_M.gguf — Marginal same-model improvement. Not worth selling as separate upgrade. Keep or delete — your choice.

14. Hardware Direction — Optional

Scout works fully without hardware. Reference hardware: KEYESTUDIO Mini Tank Kit V2 (Patrick already owns one). Tank-style chassis, motorized treads, ultrasonic sensors, 8x16 LED panel, Bluetooth. Hardware Mode is opt-in. Not every Scout owner wants hardware. The \$9.99 baseline must work fully on the phone alone.

15. Future Vision — Scout Writes His Own Behavioral Code

A long-term goal. Scout notices patterns in user behavior, generates a suggestion for a new behavior, and asks permission before activating it. Nothing changes without the user's explicit consent. Proposal Sandbox layer and approval workflow already designed. Scout cannot write compiled Kotlin code — but CAN generate behavioral scripts, response patterns, routing rules, and habit triggers within the existing safe framework.